



Tennessee Valley Authority, Post Office Box 2000, Decatur, Alabama 35609-2000

April 14, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

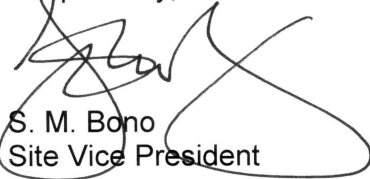
Browns Ferry Nuclear Plant, Unit 2
Renewed Facility Operating License No. DPR-52
NRC Docket No. 50-260

Subject: **Licensee Event Report 50-260/2017-001-00**

The enclosed Licensee Event Report provides details a spurious High Pressure Coolant Injection Safety System Functional Failure due to a blown fuse. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact M. W. Oliver, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,



S. M. Bono
Site Vice President

Enclosure: Licensee Event Report 50-260/2017-001-00 – High Pressure Coolant Injection Safety System Functional Failure Due to a Blown Fuse.

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 2**

Licensee Event Report 50-260/2017-001-00

High Pressure Coolant Injection Safety System Functional Failure Due to a Blown Fuse

See Enclosed

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104			EXPIRES: 03/31/2020			
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block)					Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.						
1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 2					2. DOCKET NUMBER 05000-260			3. PAGE 1 OF 6			
4. TITLE High Pressure Coolant Injection Safety System Functional Failure Due to a Blown Fuse											
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED		
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER	
02	16	2017	2017	- 001	- 00	04	14	2017	N/A	N/A	
									FACILITY NAME	DOCKET NUMBER	
									N/A	N/A	
9. OPERATING MODE 1		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)			☐ 50.73(a)(2)(viii)(A)
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)			☐ 50.73(a)(2)(viii)(B)
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)			☐ 50.73(a)(2)(ix)(A)
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)			☐ 50.73(a)(2)(x)
10. POWER LEVEL 076		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)			☐ 73.71(a)(4)
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)			☐ 73.71(a)(5)
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)			☐ 73.77(a)(1)
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)			☐ 73.77(a)(2)(i)
		☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)			☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)			☐ OTHER			Specify in Abstract below or in NRC Form 366A			
12. LICENSEE CONTACT FOR THIS LER											
LICENSEE CONTACT Ryan Coons, Licensing Engineer								TELEPHONE NUMBER (Include Area Code) 256-729-2070			
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT											
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX		
X	BJ	FU	S156	Y	N/A	N/A	N/A	N/A	N/A		
14. SUPPLEMENTAL REPORT EXPECTED ☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO					15. EXPECTED SUBMISSION DATE			MONTH	DAY	YEAR	
								N/A	N/A	N/A	
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On February 16, 2017, the spurious failure of a fuse protecting a High Pressure Coolant Injection (HPCI) system flow controller rendered the HPCI system inoperable. Operators replaced both the line and neutral fuses, and restored HPCI availability. Following a period of monitoring the current flow through the fuse and HPCI system operation tests, Operations declared the Unit 2 HPCI system to be Operable on February 17, 2017. Since HPCI is a single-train safety system, any period of unplanned inoperability constitutes a safety-system functional failure affecting accident mitigation, and is reportable. However, in the event of an emergency, the Reactor Core Isolation Cooling (RCIC) system remained operable, and all other Emergency Core Cooling Systems and the Automatic Depressurization System were available throughout this event to facilitate core cooling. Failure analysis indicates that the fuse failed when its internal resistor lead and its tension/retraction spring became uncoupled at their soldered junction, as a result of age-induced solder creep. Corrective Actions include the prompt replacement of the failed fuses, determining the population of fuses on the HPCI system and RCIC system that should be replaced on a one-time basis, and to initiate work orders to replace these fuses.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Browns Ferry Nuclear Plant, Unit 2		05000-260		YEAR	SEQUENTIAL NUMBER	REV NO.
				2017	- 001	- 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Unit 2, was in Mode 1 at approximately 76 percent power.

II. Description of Event**A. Event Summary**

On February 16, 2017, at 1052 Central Standard Time (CST), while the BFN Unit 2 High Pressure Coolant Injection (HPCI) system [BJ] was in standby readiness, Operations received the HPCI 120VAC POWER FAILURE alarm on Panel 2-9-3F. The crew responded per the Alarm Response Procedure and identified that fuse [FU] 2-FU2-073-0033C in the Auxiliary Instrument Room had cleared. This fuse protected the HPCI system flow controller (2-FIC-77-33), power supplies to the governor controls, and power supplies to various pressure transmitters from overcurrent events. Since the HPCI system is a single-train safety system, any period of unplanned inoperability constitutes a safety-system functional failure affecting accident mitigation, and is a reportable event.

Operators replaced both the line and neutral fuses, and HPCI system availability was restored on February 16, 2017 at 1145 CST. Following a period of monitoring the current flow through the fuse and HPCI system operation tests, Operations declared the Unit 2 HPCI system Operable on February 17, 2017 at 1730 CST.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components whose inoperability contributed to this event.



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C. Dates and approximate times of occurrences

Dates & Approximate Times

February 16, 2017,
at 1052 CST

Occurrence

Operations received a HPCI 120VAC POWER FAILURE alarm.

HPCI system is declared inoperable due to the loss of 120 VAC power to the HPCI Flow Controller, 2-FIC-73-33.

The Unit Supervisor immediately verified Reactor Core Isolation Cooling (RCIC) system [BN] operability by administrative means, in accordance with the Required Actions for Technical Specification (TS) Limiting Condition for Operation 3.5.1, Condition C.

February 16, 2017,
at 1145 CST

Operations declared the Unit 2 HPCI system available following fuse replacement for the HPCI system flow controller power supply. The Unit 2 HPCI system still remained inoperable pending further investigation.

February 17, 2017,
at 1730 CST

Operations declared the Unit 2 HPCI system Operable following fuse replacement and post-maintenance testing.

D. Manufacturer and model number of each component that failed during the event

The failed component was a Shawmut Company fuse, model number AJT-3.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected by this event.

F. Method of discovery of each component or system failure or procedural error

Failure was discovered on February 16, 2017, at 1052 CST when the 120 VAC power supply circuit for HPCI system flow controller fuse cleared, triggering the HPCI 120VAC POWER FAILURE alarm on Control Room Panel 2-9-3F.

G. The failure mode, mechanism, and effect of each failed component

The fuse failed when its internal resistor lead and its tension/retraction spring became uncoupled at their soldered junction. Analysis by the Tennessee Valley Authority (TVA) Central Laboratories Services determined that this was a spontaneous failure which was not due to an overcurrent event. The presumed failure mechanism is solder creep, which is a time-dependent failure mechanism.

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CONTINUATION SHEET**

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The exact age and installation date of the failed fuse was unable to be determined. Based on the history of the name brand, the failed fuse was manufactured sometime between 1976 and 1999.

H. Operator actions

There were no operator actions associated with this event.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event**A. Cause of each component or system failure or personnel error**

The fuse failed when its internal resistor lead and its tension/retraction spring became uncoupled at their soldered junction, which was presumably weakened by solder creep.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

TVA is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. The condition was discovered on February 16, 2017, when the spurious failure of the fuse protecting the HPCI system flow controller signaled an alarm [FA] in the Control Room [NA]. This loss of power rendered Unit 2 HPCI system inoperable. Since HPCI system is a single-train safety system, the unplanned inoperability caused by the fuse failure constitutes a reportable event, which is considered a safety system functional failure, and a reportable event.

V. Assessment of Safety Consequences

This event resulted in inoperability and unavailability of the single train of the BFN Unit 2 HPCI system resulting in the inability of the HPCI system to perform its safety function. In the event of an emergency, the RCIC system remained operable, and all other Emergency Core Cooling Systems (ECCS) and Automatic Depressurization System (ADS) were available during this event to facilitate core cooling.

Based on the discussion above, during the time period that the HPCI system was inoperable, sufficient systems were available to provide the required safety functions to protect the health and safety of the public.



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A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The TS Required Actions and Completion Times for HPCI system inoperability is to immediately verify that RCIC system is operable by administrative means, and restore HPCI system operability within 14 days. If these actions do not meet their Completion Times, then TS requires Unit 2 to enter Mode 3 in 12 hours. Operations personnel immediately verified that RCIC was operable by administrative means on February 16, 2017 at 1052 CST. During this event, all other ECCS, including the ADS, were available to mitigate abnormal and accident conditions.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

The HPCI system was determined to be inoperable when a blown fuse resulted in a HPCI 120VAC POWER FAILURE alarm actuation on February 16, 2017 at 1052 CST. The HPCI system was declared operable on February 17, 2017 at 1730 CST following post-maintenance testing. Approximately 31 hours elapsed between the time of discovery and restoring operability.

VI. Corrective Actions

Corrective Actions are being managed by TVA's corrective action program under Condition Report (CR) 1263268.

A. Immediate Corrective Actions

The Operations crew responded in accordance with the Alarm Response Procedure, and identified that fuse 2-FU2-073-0033C in the Auxiliary Instrument Room had cleared. Operators promptly replaced both the line and neutral fuses to restore HPCI functionality. The Unit 2 HPCI system was declared available by Operations within 1 hour of fuse failure.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

Corrective Actions included the prompt replacement of the failed fuses, a determination of the population of HPCI system and RCIC system fuses that should be replaced on a one-time basis, and the replacement of those fuses.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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VII. Previous Similar Events at the Same Site

In May 2002, the NRC published NUREG-1760, an aging assessment of safety-related fuses. This study found that fusible link degradation accounts for approximately 80% of fuse failure events occurring below the fuse's rated current.

A search of the BFN Corrective Action Program identified 12 fuses which spuriously opened since 2010. The search did not identify a fuse failure at the location described in this LER.

A review of BFN LERs for the last five years identified one event where a blown fuse resulted in an actual safety system functional failure: LER 50-260/2016-001-01, High Pressure Coolant Injection Safety System Functional Failure due to a Blown Fuse and a Failed Relay. For this event, the fuse failure was due to test equipment, not aging effects. Therefore, the corrective actions for this previous event would not have prevented the current event.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.